

Foundation Models of Electronic Health Records

ELLIS Summer School 2026

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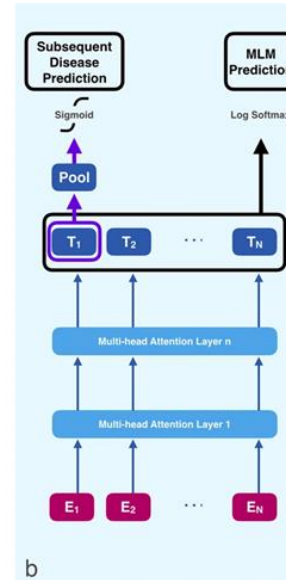
BEHRT

Transformer for EHR



a

Li Y. et al. **Scientific Reports** 2020.

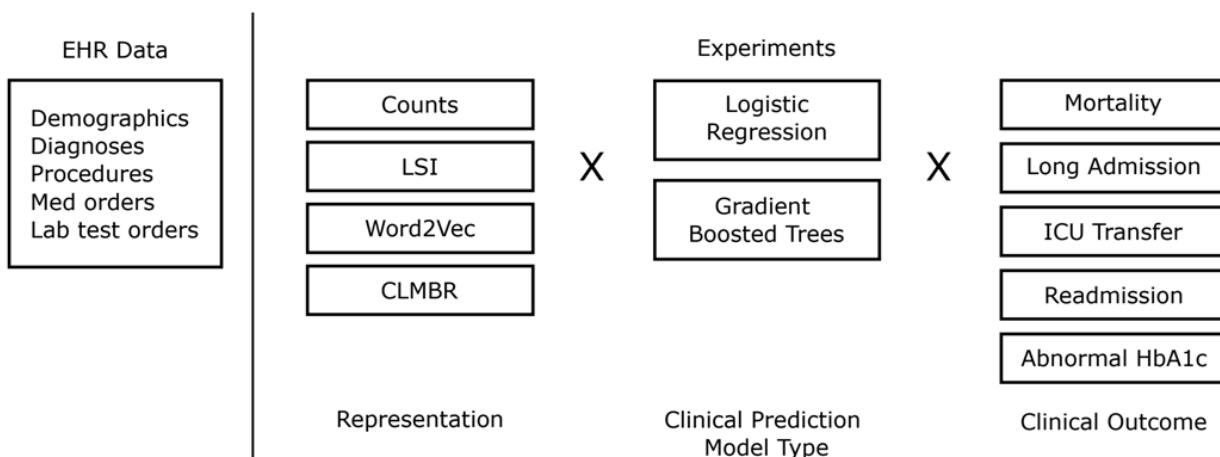


b



CLMBR

Language Models An Effective Patient Representation Learning Technique For EHRs

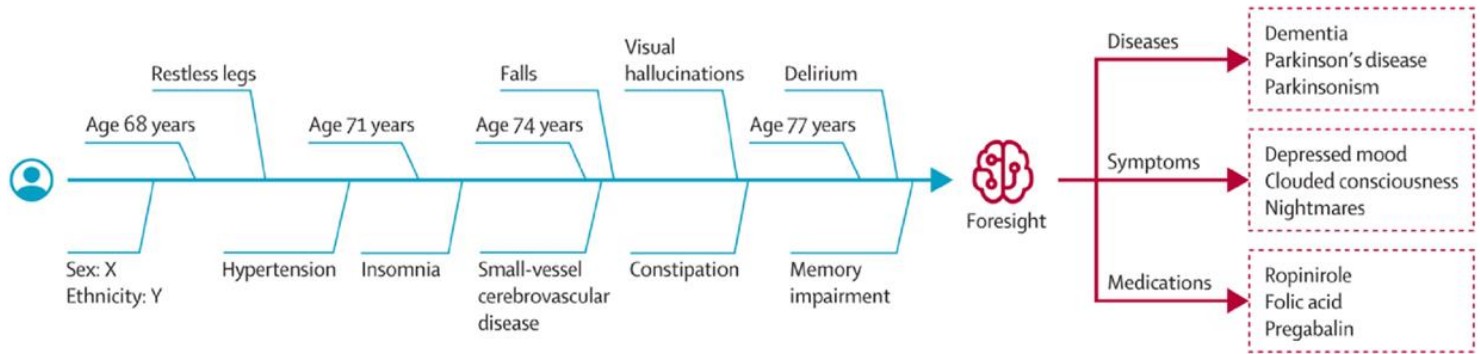


Steinberg E. et al. **arXiv** 2020.



Foresight

Learning from past patient data, to predict the future

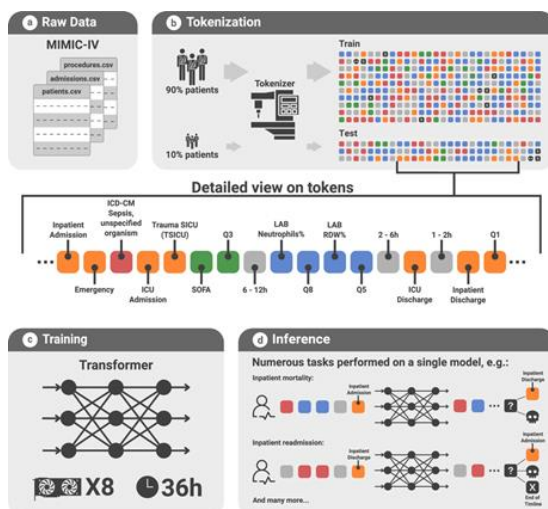


Kraljevic Z. et al. **arXiv 2022 / Lancet Digital Health 2024.**



ETHOS

Zero shot health trajectory prediction using transformer

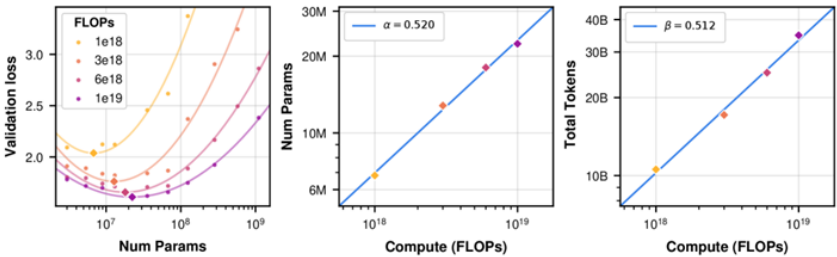
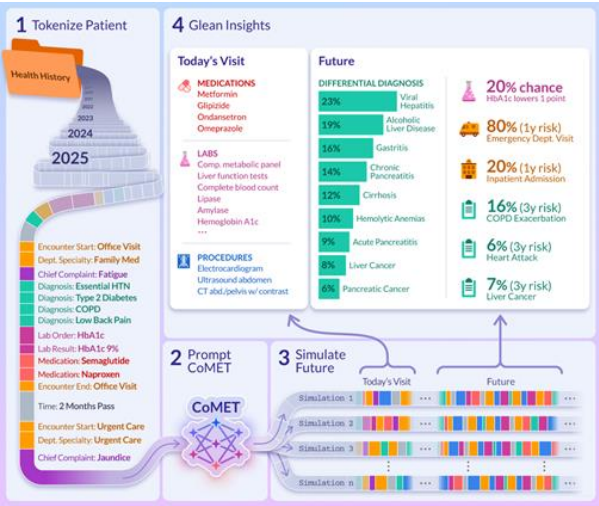


Renc P. et al. **npj Digital Medicine 2024.**



CoMET/Curiosity

Generative Medical Event Models Improve with Scale



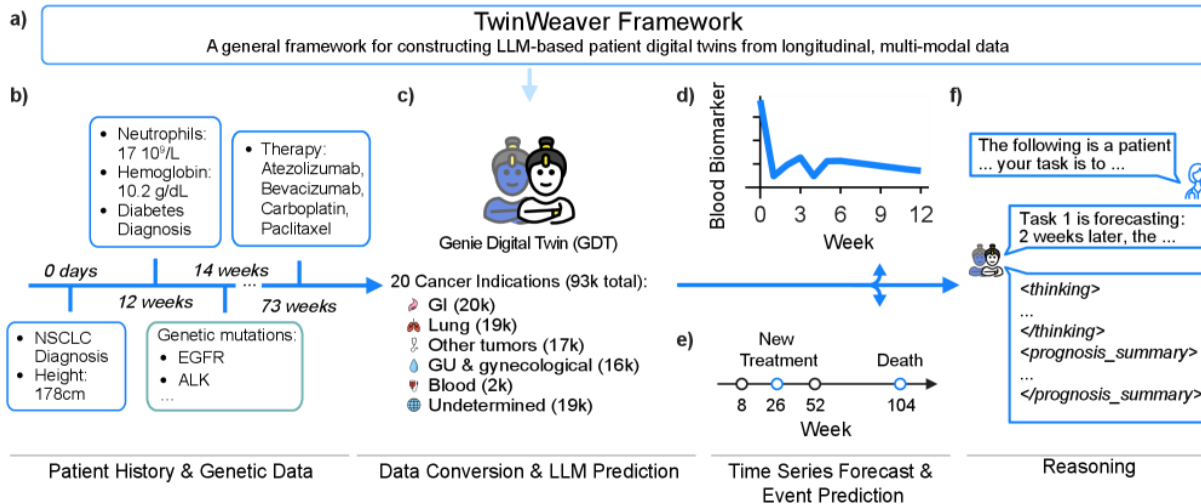
Waxler S. et al. **arXiv 2025**.

Innovations in LLMs

Stage/Innovation	LLM	FEMR
Pretraining (2019)	GPT-2 -> Complete a paragraph	Foresight/ETHOS/Curiosity/.. -> Predict patient trajectory
Instruction Tuning/RLHF (2022)	ChatGPT -> Answer questions	???
RLVR (2024)	GPT-o1 -> Agentic coding/Maths Proofs	???

TwinWeaver

An LLM-Based Foundation Model Framework for Pan-Cancer Digital Twins



Makarov N. et al. **arXiv 2026**.

Honorable Mentions

Reading for the journey home..

- **MOTOR**. Steinberg E. et al. arXiv 2023
 - Time-to-event pretraining
- **Delphi-2M**. Shmatko A. et al. Nature 2025.
 - Cross entropy -> Hazard
 - External validation: UK Biobank -> Denmark
- **Traj-CoA**. Zheng S. et al. arXiv 2025
 - Agentic approach
- **SMB-Structure**. Adam I. et al. arXiv 2026.
 - SFT + JEPA loss
- **APOLLO**. Zhang A. et al. arXiv 2026.
 - Multimodal integration



Part of the
NHS Research Secure Data
Environment Network



British Heart Foundation
Data Science Centre
Led by Health Data Research UK



Foresight-England

A national-scale foundation model of electronic health records, for COVID-19 research

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